Drag Drop Questions 3

8 questions | Original English questions and source answers

Question 1

Refer to the exhibit. An engineer is required to verify that the network parameters are valid for the users wireless LAN connectivity on a /24 subnet. Drag and drop the values from the left onto the network parameters on the right. Not all values are used.



Available items

192.168.1.20

192.168.1.255

192.168.1.0

192.168.1.1

B8-76-3F-7C-57-DF

192.168.1.254

1A-76-3F-7C-57-DF

Descriptions / target positions

host IP address

broadcast address

network address

default gateway

MAC address

last assignable IP address in the subnet

Unused

Answer

Description / position	Source answer
host IP address	192.168.1.20
broadcast address	192.168.1.255
network address	192.168.1.0
default gateway	192.168.1.1
MAC address	B8-76-3F-7C-57-DF
last assignable IP address in the subnet	192.168.1.254
Unused	1A-76-3F-7C-57-DF

Source explanation

+ host IP address: 192.168.1.20
+ broadcast address: 192.168.1.255
+ network address: 192.168.1.0
+ default gateway: 192.168.1.1
+ MAC address: B8-76-3F-7C-57-DF
+ last assignable IP address in the subnet: 192.168.1.254

Source: CCNA Training » Drag Drop Quiz 3.pdf | Question 1

Question 2

Drag and drop the SNMP components from the left onto the descriptions on the right.

Available items

SNMP trap

SNMP agent

MIB

SNMP manager

Descriptions / target positions

unsolicited message

responds to status requests and requests for information about a device

collection of variables that can be monitored

resides on an NMS

Answer

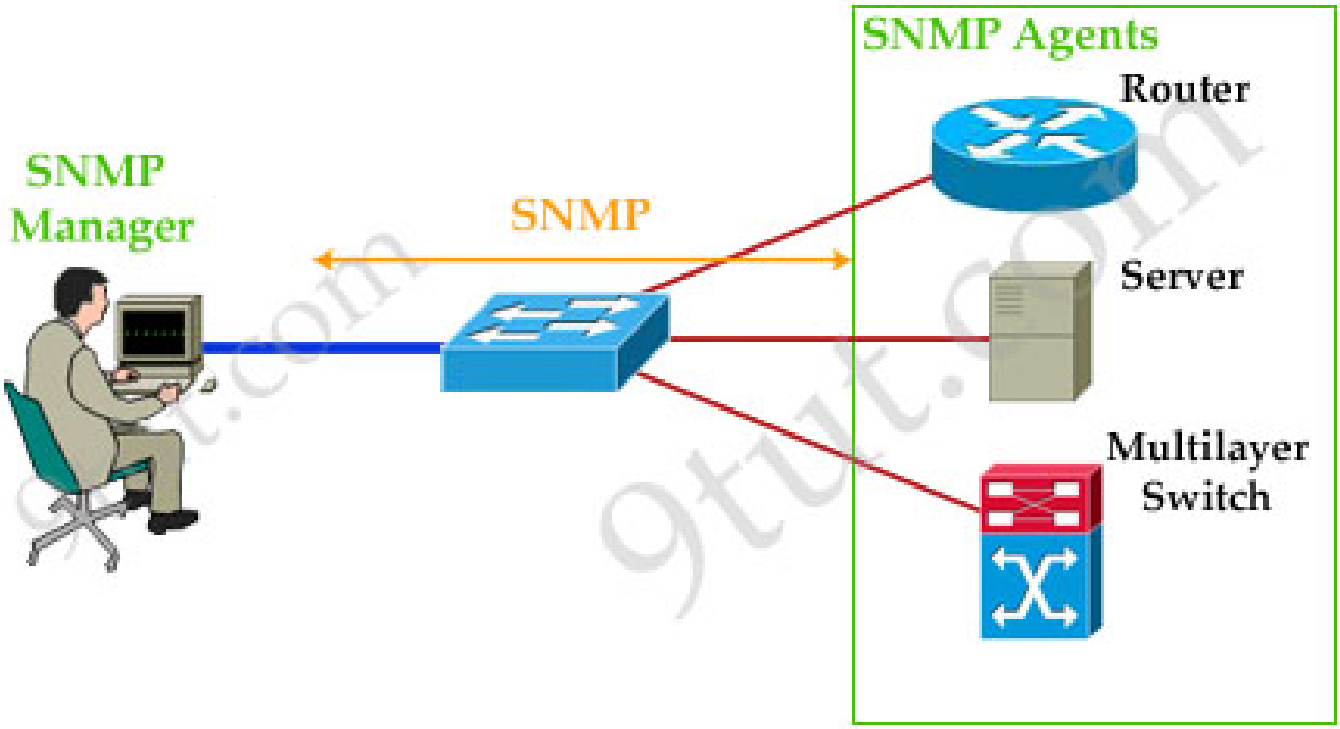
Description / position	Source answer
unsolicited message	SNMP trap
responds to status requests and requests for information about a device	SNMP agent
collection of variables that can be monitored	MIB
resides on an NMS	SNMP manager

Source explanation

+ unsolicited message: SNMP trap
+ responds to status requests and requests for information about a device: SNMP agent
+ collection of variables that can be monitored: MIB
+ resides on an NMS: SNMP manager

SNMP consists of 3 items:

+ SNMP Manager (sometimes called Network Management System – NMS): a software runs on the device of the network administrator (in most case, a computer) to monitor the network.
+ SNMP Agent: a software runs on network devices that we want to monitor (router, switch, server…)
+ Management Information Base (MIB): is the collection of managed objects. This components makes sure that the data exchange between the manager and the agent remains structured. In other words, MIB contains a
set of questions that the SNMP Manager can ask the Agent (and the Agent can understand them). MIB is commonly shared between the Agent and Manager.



Source: CCNA Training » Drag Drop Quiz 3.pdf | Question 2

Question 3

Drag and drop the QoS congestion management terms from the left onto the description on the right.

Available items

CQ

CBWFQ

PQ

WFQ

FIFO

Descriptions / target positions

services a specified number of bytes in one queue before continuing to the next queue

provides guaranteed bandwidth to a specified class of traffic

places packets into one of four priority-based queues

provides minimum guaranteed bandwidth to one or more flows

uses store-and-forward queuing

Answer

Description / position	Source answer
services a specified number of bytes in one queue before continuing to the next queue	CQ
provides guaranteed bandwidth to a specified class of traffic	CBWFQ
places packets into one of four priority-based queues	PQ
provides minimum guaranteed bandwidth to one or more flows	WFQ
uses store-and-forward queuing	FIFO

Source explanation

+ services a specified number of bytes in one queue before continuing to the next queue: CQ
+ provides guaranteed bandwidth to a specified class of traffic: CBWFQ
+ places packets into one of four priority-based queues: PQ
+ provides minimum guaranteed bandwidth to one or more flows: WFQ
+ uses store-and-forward queuing: FIFO
First-in, first-out (FIFO): FIFO entails no concept of priority or classes of traffic. With FIFO, transmission of packets out the interface occurs in the order the packets arrive, which means no QoS
Priority Queuing (PQ): schedules traffic such that the higher-priority queues “always” get serviced first
Custom Queuing (CQ): provide specific traffic guaranteed bandwidth at a potential congestion point, assuring the traffic a fixed portion of available bandwidth and leaving the remaining bandwidth to other traffic.
Weighted fair queueing (WFQ): offers dynamic, fair queuing that divides bandwidth across queues of traffic based on weights. In standard WFQ, packets are classified into flows according to one of four criteria: the source Internet Protocol address (IP address), the destination IP address, the source Transmission Control Protocol (TCP) or User Datagram Protocol (UDP) port, or the destination TCP or UDP port.
Class-based weighted fair queueing (CBWFQ) extends the standard WFQ functionality to provide support for user-defined traffic classes. For CBWFQ, you define traffic classes based on match criteria including protocols, access control lists (ACLs), and input interfaces. Packets satisfying the match criteria for a class constitute the traffic for that class. A queue is reserved for each class, and traffic belonging to a class is directed to the queue
for that class.
Once a class has been defined according to its match criteria, you can assign it characteristics. To characterize a
class, you assign it bandwidth, weight, and maximum packet limit. The bandwidth assigned to a class is the guaranteed bandwidth delivered to the class during congestion.

Source: CCNA Training » Drag Drop Quiz 3.pdf | Question 3

Question 4

An engineer is tasked to configure a switch with port security to ensure devices that forward unicasts, multicasts, and broadcasts are unable to flood the port. The port must be configured to permit only two random MAC addresses at a time. Drag and drop the required configuration commands from the left onto the sequence on the right. (Not all commands are used)

Available items

```
switchport mode access
switchport port-security
switchport port-security mac-address 0060.3EDD.77ab
switchport port-security mac-address 00D0.D443.4564
switchport port-security mac-address sticky
switchport port-security violation shutdown
switchport port-security maximum 2
```

Descriptions / target positions

1

2

3

4

Answer

Description / position	Source answer
1	switchport mode access
2	switchport port-security
3	switchport port-security maximum 2
4	switchport port-security violation shutdown

Unused items

switchport port-security mac-address 0060.3EDD.77ab

switchport port-security mac-address 00D0.D443.4564

switchport port-security mac-address sticky

Source explanation

This question says "permit only two random MAC addresses at a time" so we cannot use the "switchport port-security mac-address sticky" command as the switch will only learn and save the first two MAC addresses forever. Although the command "switchport port-security violation shutdown" is unnecessary because it is the default state of port-security but it is not wrong to reuse it.
Note:
The full syntax of the second command is:
switchport port-security mac-address sticky [MAC]
If we don’t specify the MAC address (like in this question) then the switch will dynamically learn the attached MAC Address and place it into your running-configuration.

Source: CCNA Training » Drag Drop Quiz 3.pdf | Question 4

Question 5

Drag the characteristics of network architectures from the left onto the type of architecture on the right.

Available items

Collapsed Core

Three-Tier

Descriptions / target positions

most appropriate for small network designs

single device handles the core and the distribution layer

separate devices handle the core and the distribution layer

more cost-effective than other options

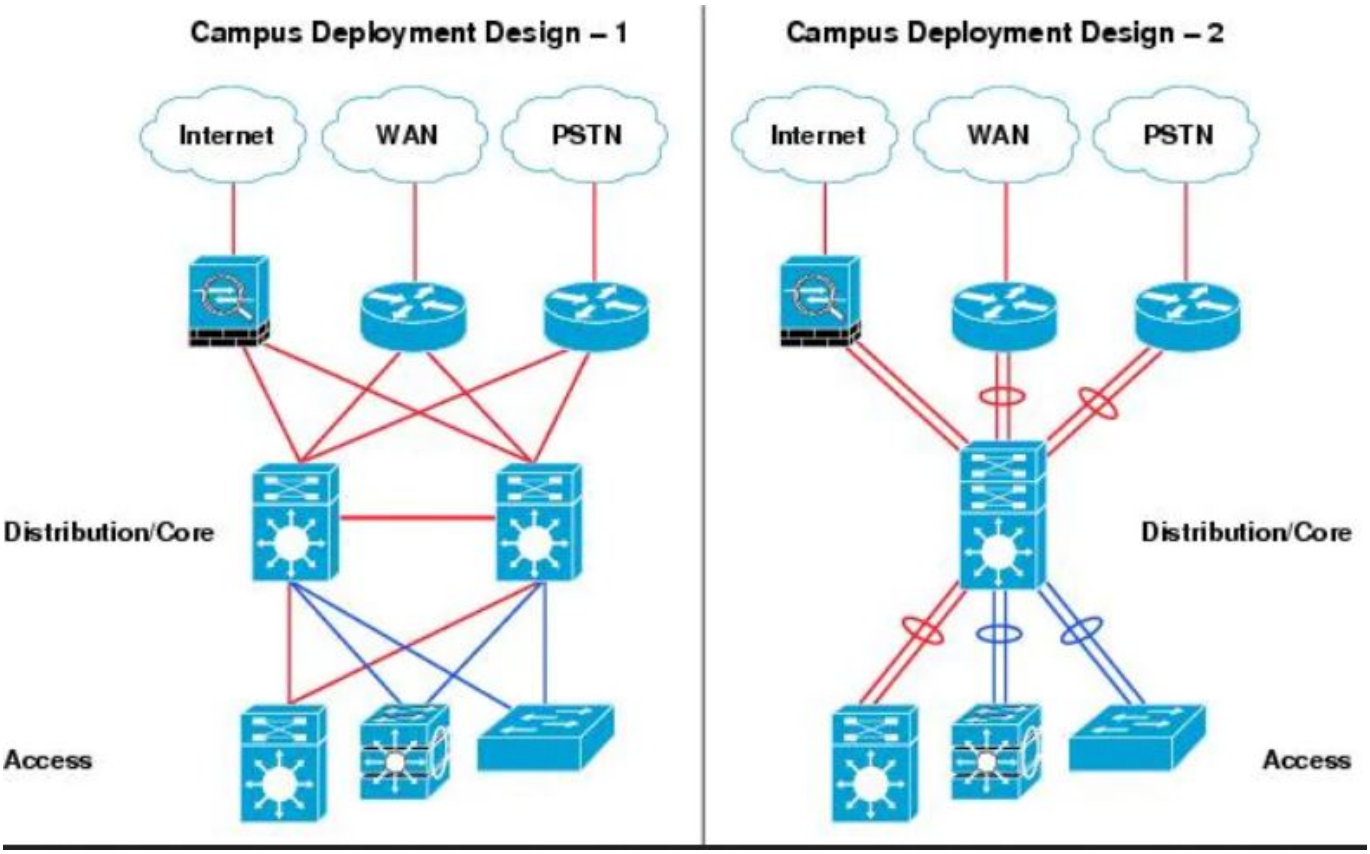
enhances network availability

Answer

Description / position	Source answer
most appropriate for small network designs	Collapsed Core
single device handles the core and the distribution layer	Collapsed Core
separate devices handle the core and the distribution layer	Three-Tier
more cost-effective than other options	Collapsed Core
enhances network availability	Three-Tier

Source explanation

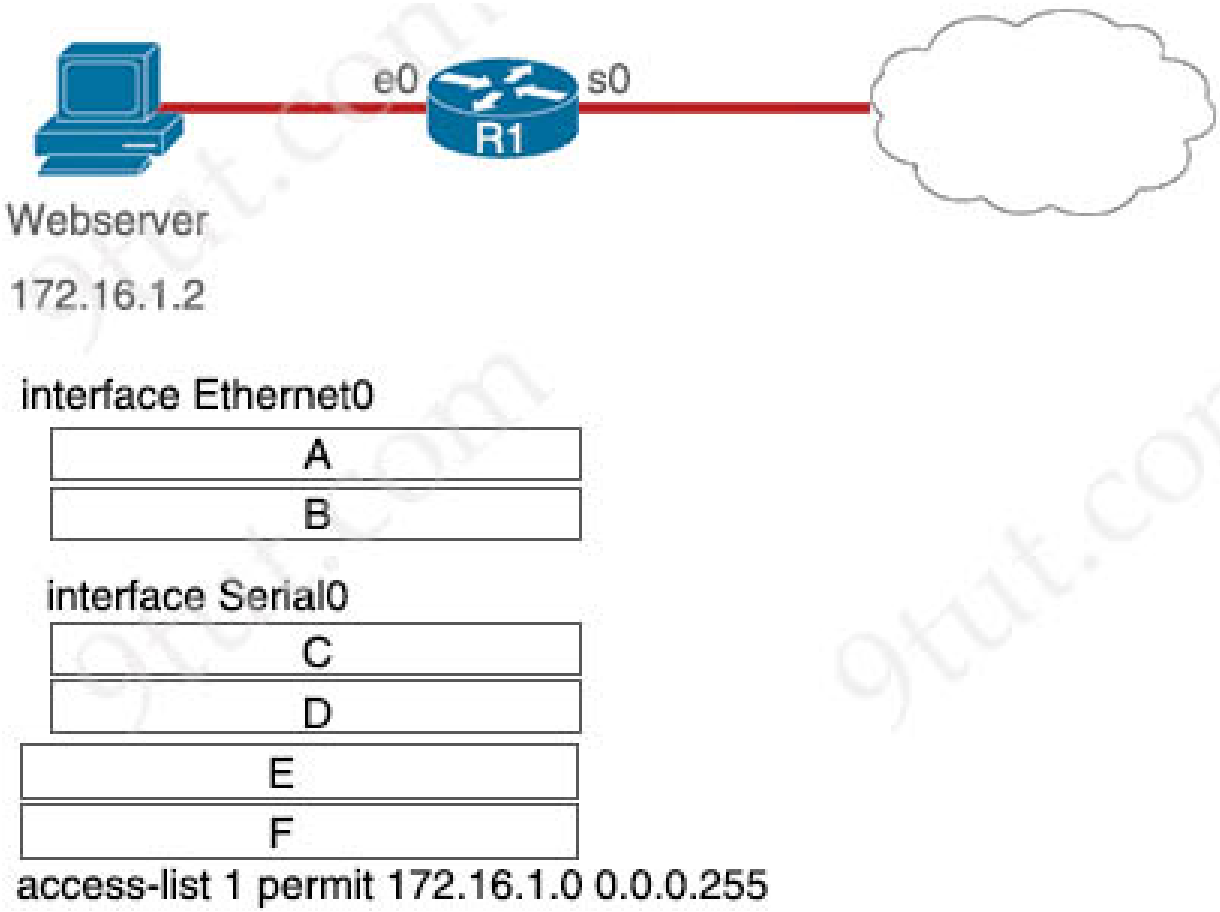
The three-tier hierarchical design maximizes performance, network availability, and the ability to scale the network design.
However, many small enterprise networks do not grow significantly larger over time. Therefore, a two-tier hierarchical design where the core and distribution layers are collapsed into one layer is often more practical.
A
“collapsed core” is when the distribution layer and core layer functions are implemented by a single device. The primary motivation for the collapsed core design is reducing network cost, while maintaining most of the benefits of the three-tier hierarchical model.
Reference: https://www.ciscopress.com/articles/article.asp?p=2202410&seqNum=4
A collapsed core network is shown below. The collapsed core network may be deployed with redundant core/distribution router, or consolidated core/distribution router.



Source: CCNA Training » Drag Drop Quiz 3.pdf | Question 5

Question 6

Refer to the exhibit. An engineer is configuring the router to provide static NAT for the webserver. Drag and drop the configuration commands from the left onto the letters that correspond to its position in the configuration on the right.



Available items

```
ip address 172.16.1.1 255.255.255.0
ip nat inside
ip address 45.83.2.214 255.255.255.240
ip nat outside
ip nat inside source static tcp 172.16.1.2 80 45.83.2.214 80 extendable
ip nat inside source list 1 interface s0 overload
```

Descriptions / target positions

position A

position B

position C

position D

position E

position F

Answer

Description / position	Source answer
position A	ip address 172.16.1.1 255.255.255.0
position B	ip nat inside
position C	ip address 45.83.2.214 255.255.255.240
position D	ip nat outside
position E	ip nat inside source static tcp 172.16.1.2 80 45.83.2.214 80 extendable
position F	ip nat inside source list 1 interface s0 overload

Source explanation

+ position A: ip address 172.16.1.1 255.255.255.0
+ position B: ip nat inside
+ position C: ip address 45.83.2.214 255.255.255.240
+ position D: ip nat outside
+ position E: ip nat inside source static tcp 172.16.1.2 80 45.83.2.214 80 extendable
+ position F: ip nat inside source list 1 interface s0 overload

Source: CCNA Training » Drag Drop Quiz 3.pdf | Question 6

Question 7

Drag the IPv6 DNS record types from the left onto the description on the right.

Available items

NS

SOA

CNAME

PTR

AAAA

Descriptions / target positions

correlates a domain with its authoritative name servers

associates the domain serial number with its owner

aliases one name to another

supports reverse name lookups

correlates a host name with an IP address

Answer

Description / position	Source answer
correlates a domain with its authoritative name servers	NS
associates the domain serial number with its owner	SOA
aliases one name to another	CNAME
supports reverse name lookups	PTR
correlates a host name with an IP address	AAAA

Source explanation

+ correlates a domain with its authoritative name servers: NS
+ associates the domain serial number with its owner: SOA
+ aliases one name to another: CNAME
+ supports reverse name lookups: PTR
+ correlates a host name with an IP address: AAAA
An AAAA record maps a domain name to the IP address (Version 6) of the computer hosting the domain. An AAAA record is used to find the IP address of a computer connected to the internet from a name. The AAAA record is conceptually similar to the A record, but it allows you to specify the IPv6 address of the server, rather than the IPv4. An example of AAAA is the 'www' of 9tut.com is pointed to
2001:0db8:aaaa:bbbb:cccc:dddd:eeee:ffff.
NS (name server) looks like a website URL, but instead of linking to a website, they link to the domain names authoritative nameservers. An example of a NS is 'ns1.9tut.com'.
The Canonical Name record (CNAME) tells anyone visiting that subdomain to use the same DNS records as another domain / subdomain. If you are already using an A or AAAA record for a subdomain you wouldn't use a CNAME. CNAME records must always point to another domain / subdomain, never directly to an IP address. For example, we can configure the CNAME of www.9tut.com points to 9tut.com so that they use the same IP address.
As opposed to forward DNS resolution (A and AAAA DNS records), the Reverse-lookup Pointer (PTR) record is used to look up domain names based on an IP address.
An SOA record or start of authority record specifies the DNS server providing authoritative information about an Internet domain, the email of the domain administrator, the domain serial number, and several timers relating
to refreshing the zone.
An example of a SOA record is shown below.
; name TTL class rr Nameserver email-address
mydomain.com. 14400 IN SOA ns.mynameserver.com. root.ns.mynameserver.com. (
2004123001 ; Serial number
86000 ; Refresh rate in seconds
7200 ; Update Retry in seconds
3600000 ; Expiry in seconds
600 ; minimum in seconds)
name – mydomain.com is the main name in this zone.
TTL – 14400 – TTL defines the duration in seconds that the record may be cached by client side programs. If it is set as 0, it indicates that the record should not be cached. The range is defined to be between 0 to 2147483647
(close to 68 years !) .
Class – IN – The class shows the type of record. IN equates to Internet. Other options are all historic. So as long as your DNS is on the Internet or Intranet, you must use IN.
Nameserver – ns.nameserver.com. – The nameserver is the server which holds the zone files. It can be either an
external server in which case, the entire domain name must be specified followed by a dot. In case it is defined
in this zone file, then it can be written as "ns" .
Email address – root.ns.nameserver.com. – This is the email of the domain name administrator. Now, this is really confusing, because people expect an @ to be in an email address. However in this case, email is sent to root@ns.nameserver.com, but written as root.ns.nameserver.com . And yes, remember to put the dot behind the
domain name.
Serial number – 2004123001 – This is a sort of a revision numbering system to show the changes made to the DNS Zone. This number has to increment , whenever any change is made to the Zone file. The standard convention is to use the date of update YYYYMMDDnn, where nn is a revision number in case more than one updates are done in a day. So if the first update done today would be 2005301200 and second update would be
2005301201.
Refresh – 86000 – This is time(in seconds) when the slave DNS server will refresh from the master. This value represents how often a secondary will poll the primary server to see if the serial number for the zone has increased (so it knows to request a new copy of the data for the zone). It can be written as "23h88M" indicating
23 hours and 88 minutes. If you have a regular Internet server, you can keep it between 6 to 24 hours.

Retry – 7200 – Now assume that a slave tried to contact the master server and failed to contact it because it was
down. The Retry value (time in seconds) will tell it when to get back. This value is not very important and can be a fraction of the refresh value.
Expiry – 3600000 – This is the time (in seconds) that a slave server will keep a cached zone file as valid, if it can’t contact the primary server. If this value were set to say 2 weeks (in seconds), what it means is that a slave
would still be able to give out domain information from its cached zone file for 2 weeks, without anyone knowing the difference. The recommended value is between 2 to 4 weeks.
Minimum – 600 – This is the default time(in seconds) that the slave servers should cache the Zone file. This is the most important time field in the SOA Record. If your DNS information keeps changing, keep it down to a day or less. Otherwise if your DNS record doesn’t change regularly, step it up between 1 to 5 days. The benefit of keeping this value high, is that your website speeds increase drastically as a result of reduced lookups. Caching servers around the globe would cache your records and this improves site performance.

Source: CCNA Training » Drag Drop Quiz 3.pdf | Question 7

Question 8

Drag and drop the DHCP snooping terms from the left onto the descriptions on the right.

Available items

snooping binding database

spurious DHCP server

DHCP server

untrusted

trusted

Descriptions / target positions

list of hosts on the network that are unknown to the administrative domain

unknown DHCP server within an administrative domain

network component that propagates IP addresses to hosts on the network

default state of all interfaces

internal device under the control of the network administrator

Answer

Description / position	Source answer
list of hosts on the network that are unknown to the administrative domain	snooping binding database
unknown DHCP server within an administrative domain	spurious DHCP server
network component that propagates IP addresses to hosts on the network	DHCP server
default state of all interfaces	untrusted
internal device under the control of the network administrator	trusted

Source explanation

+ list of hosts on the network that are unknown to the administrative domain: snooping binding database
+ unknown DHCP server within an administrative domain: spurious DHCP server
+ network component that propagates IP addresses to hosts on the network: DHCP server
+ default state of all interfaces: untrusted
+ internal device under the control of the network administrator: trusted

Source: CCNA Training » Drag Drop Quiz 3.pdf | Question 8